# Weihsuan Wong

**Staff Software Engineer | AI Developer Experience & UX Engineering**

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## SUMMARY

Staff Software Engineer with 6+ years building **AI-powered developer tools**, **frontend platforms**, and **accessible product experiences** at Visa and ASUS. Combine hands-on React/TypeScript architecture with design engineering, rapid experimentation, and developer-centered workflows. Built core collaboration and state-management experiences for Visa Blitz Code, an AI prototyping platform adopted by 5,000+ employees, generating 6,000+ prototypes and 10M+ lines of code in two months.

## WORK EXPERIENCE

**VISA - CURRENCYCLOUD** | Austin, TX

**Staff SW Engineer - Front-End Developer** Jun 2026 - Present

- **Lead frontend platform architecture for Backoffice 2.0**, establishing micro-frontend integration contracts, shared host patterns, and module ownership and release standards across product teams.
- **Develop Cosmos Code, an internal VS Code AI assistant**, and workflows for module scaffolding, UI generation, validation, and developer onboarding, with human approval integrated into the developer experience.
- **Collaborate with Product, Design, Security, and Infrastructure** on secure module delivery, authentication boundaries, and design-system adoption to make developer experiences reliable and production-ready.

**VISA - USER EXPERIENCE** | Austin, TX

**Senior Design Engineer** Feb 2025 - Jun 2026

- **Designed and implemented the end-to-end collaboration and permission system** for Visa Blitz Code (AI prototyping platform, 5,000+ employees, 10M+ lines generated), including a backend permission resolution engine aggregating multi-source access (direct / folder / nested shares) and a frontend "effective role" abstraction with progressive disclosure
- **Refactored Blitz Code's global state store into per-project isolated state maps**, eliminating an entire class of cross-project bugs (loading conflicts, state leakage) by fixing the root cause rather than patching symptoms
- **Architected the Tap NFC demo platform** used by **2,000+ external customers at Payments Forum 2025** (also featured at Stripe Sessions and Visa Investor Day); designed a 3-layer system (NFC

integration / business logic / presentation) that enabled overnight production hotfixes without UI regressions during the live event (Visa LinkedIn post)

**Design Engineer** Aug 2021 - Feb 2025

- **Led a unified code integration system across Visa Payment Design System's React, Angular, Flutter, and CSS libraries**, powering every component page on design.visa.com; designed a priority-based multi-layer state sync (URL / session / client) to handle deep linking, persistence, and shareability without conflicts
- Drove a **60% increase in user satisfaction** for Tap to Phone accessibility through a week-long user testing session with people with disabilities, translating observed gaps into WCAG-compliant product changes (LinkedIn post)
- Built scalable frontend for fintech pilot programs (e-HKD with HSBC, Hang Seng Bank) and contributed to the public launch of Visa's open-source Design System and **Led development of Visa Digital Enablement demo** through cross-functional collaboration
- **Enhanced Visa's Flutter component library** with enterprise interaction patterns (nested / indeterminate states) and accessibility improvements for real-world usability
- Designed a shared interaction service layer for Visa's Angular component library, replacing per-component logic with a single source of truth
- **Co-invented a patent** on accessibility and inclusive design for visually impaired users (patent link)

**ASUS - Gaming Gear & Accessories** | Taipei, Taiwan

**Front-End Developer** Jul 2020 - Jun 2021

- Delivered features for Armoury Crate used by **millions of ROG & TUF users worldwide**
- Built and maintained **real-time device interfaces** including RGB lighting, touch panel, and key mapping systems
- Developed features for wireless keyboards (M601, Claymore II) including power management and real-time status monitoring

**Advantech - Industrial IoT** | Taipei, Taiwan

**Front-End Intern** Jul 2019 - Aug 2019

- Developed a water treatment management dashboard using Grafana, AngularJS, and SVG.js for real-time industrial data visualization
- Presented the solution at the Taipei International Industrial Automation Exhibition 2019
- Received Advantech Summer Intern Excellence Award (2019) for leadership and process improvements

## PROJECTS & ENTREPRENEURIAL EXPERIENCE

### Avori | AI Travel Agent

**Engineer (AI/Full-Stack)** **Apr 2026 - Present**

- Build an AI-native travel planner end to end: a Flutter app (Riverpod, go_router, Google Maps) for iOS and Android, and a NestJS/TypeScript backend on PostgreSQL (Prisma), Redis, and GCP.
- Architected a multi-stage LLM planning and repair pipeline that pairs model reasoning with deterministic checks for venue hours, routing, deduplication, and schedule feasibility, so every itinerary is verified before it reaches the user.
- Designed tiered model routing across OpenAI and Gemini, sending each pipeline stage to the cheapest model that meets its quality bar, to balance output quality, latency, and inference cost.
- Engineered real-time generation streaming over WebSocket (Socket.IO) with Redis and BullMQ background jobs; the app renders the itinerary progressively (skeleton, then day previews) and reconciles against a server snapshot after reconnects or relaunches, so interrupted sessions resume without losing progress.
- Built a conversational planning UI and a map-based itinerary editor in Flutter (reorder, swap, per-stop notes, closure alerts), with offline-aware states and local persistence (Hive).
- Built itinerary editing with user-set constraints: stops a traveller locks or books are preserved through every AI edit and repair pass, with booking and itinerary writes committed in one transaction under optimistic version guards.
- Integrated Google Places and Routes with caching and fallback logic, and built a curated city-data pipeline (automated verification, monthly staleness checks in CI) to keep recommendations current across multiple cities.

### Legislator Election Official Website

Hackathon (The F2E 5th) Nov 2023

- Built a multilingual, accessible campaign platform using Next.js and TypeScript **for public information access**

### Taiwan Travel Guide, Taiwan Bike Guide, Taiwan Bus Guide

Hackathon (The F2E 3rd) Nov 2021

- Developed **map-based travel platforms** integrating transportation and POI data for trip planning
- Implemented **real-time geospatial features** including live bus tracking, bike stations, and route visualization using Mapbox

## EDUCATION

**Johns Hopkins University** - Online Program 2026 - Present

Certificate Program in Applied Generative AI and Agentic AI

**Stevens Institute of Technology** - Hoboken, USA 2020

M.S. Computer Science

**National Kaohsiung Normal University** - Kaohsiung, TW 2016

B.S. Mathematics

## CERTIFICATIONS

**Google UX Design Certificate** - Coursera 2024

**Front-End & Back-End Website Development** - Highline College 2018

## SKILLS

**Frontend:** React, TypeScript, Next.js, Astro, React Query, Redux (Thunk, Saga), Zustand, Context API, HTML, CSS, SCSS, JavaScript, Angular, Google Maps Platform (Places, Routes), Mapbox

**Architecture & UX:** Design Systems, Accessibility (WCAG), Component Architecture, Responsive Design

**Backend:** Node.js, Express, NestJS, GraphQL, REST APIs, MongoDB, Redis, WebSockets, Firebase (Cloud Firestore, Hosting, Functions, Analytics)

**AI & Product:** LLM Orchestration, AI Model Integration (GPT, Gemini), Prompt Engineering, AI-driven Product Development

**Mobile & Cross-Platform**: React Native, Flutter, Kotlin